UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

Time Series Project

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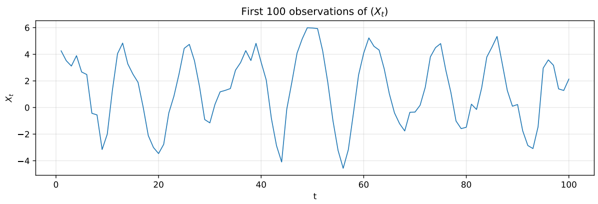
Master 1 MAEF - 2025-2026

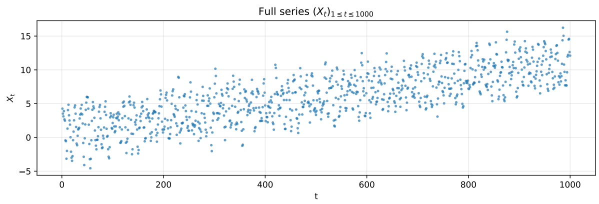
1 Descriptive analysis (EDA)

1.1 Plot of the series

Figure 1a shows the first 100 observations of (X_t) and Figure 1b the full series. Looking at the first 100 observations only, we cannot observe any tendency; however with all the observations there is clearly an upward trend. A periodic pattern is also clearly visible, of apparent period 12.



(a) X_t, $1 \leq t \leq 100$.



(b) X_t, $1 \leq t \leq 1000$.

Figure 1: Raw series. The first 100 observations alone do not reveal the trend, which becomes obvious on the full sample.

1.2 Autocorrelation function

The empirical ACF of (X_t) up to lag 40 is given in Figure 2. Three features stand out:

- **slow decay**: the ACF decreases very slowly and stays well above the confidence band for all 40 lags, suggesting a non-stationary series;
- **periodic pattern**: the ACF goes up and down, which suggests a seasonality in the series;
- **no quick drop to zero**: a stationary ARMA would have an ACF dropping to zero exponentially quickly. That is not the case here, which suggests non-stationarity. Mathematically this is consistent with the series having a unit root.

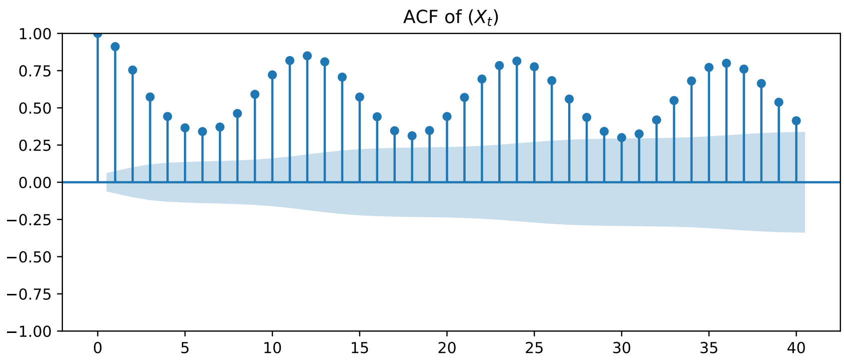
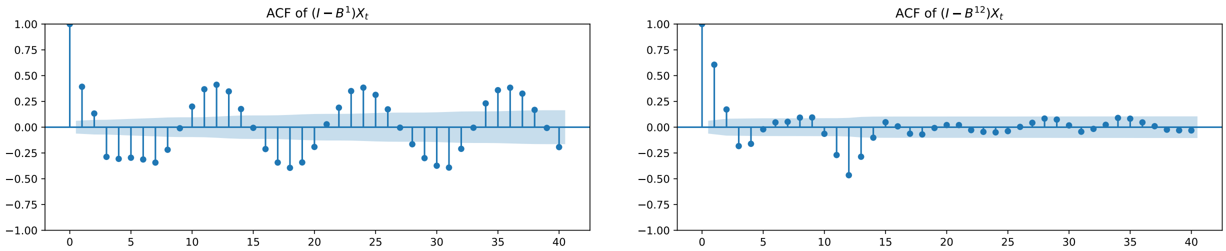


Figure 2: Empirical ACF of X_t.

Differentiation experiment. Since the ACF decays slowly, we try to differentiate $(I-B^d)X_t$ for several values of d to look for a value that produces a rapid ACF decay. Differentiating $d = 1, 2, 3, 4$ (or $d = 20, 30, \dots$) does not change the convergence speed of the ACF (Figure 3a). However differentiating with $d = 12$ (or any multiple of 12) clearly stationarises the series (Figure 3b). This is a strong indication that we are not dealing with a plain ARIMA(p, d, q) but rather with a SARIMA$(p, d, q) \times (P, D, Q)_{12}$.



(a) ACF of $(I - B)X_t$.   (b) ACF of $(I - B^{12})X_t$.

Figure 3: Differentiation does almost nothing for $d = 1$ but clearly stationarises the series for $d = 12$.

ADF test. An alternative way to see this is the Augmented Dickey–Fuller test:

- H_0: the series has a unit root (non-stationary);
- H_1: the series is stationary.

On X_t we get ADF $= -1.02$, p-value $= 0.747$. The p-value is much greater than 0.05, so we do *not* reject H_0: the series is likely non-stationary. It cannot therefore be estimated directly by an ARMA – we need an ARIMA or a SARIMA, consistent with what we observed from the ACF plots.

2 Stationarisation by estimation and modelling

We assume the additive decomposition

$$X_t = m(t) + s(t) + Y_t, \qquad E[Y_t] = 0, \quad s(t+12) = s(t), \quad \sum_{k=1}^{12} s(k) = 0,$$

where $m(t)$ is the (linear) trend and $s(t)$ is the deterministic period-12 seasonality. The brief assumes $m(t) = a + bt$.

2.1 Estimation of $\hat{m}(t)$ and $\hat{s}(t)$

Trend. The period $d = 12$ is even, so we use the standard centred moving-average estimator

$$\hat{m}(t) = \frac{1}{12}\Big(\tfrac{1}{2}X_{t-6} + X_{t-5} + \cdots + X_{t+5} + \tfrac{1}{2}X_{t+6}\Big),$$

defined for $7 \leq t \leq 994$. This filter eliminates a deterministic component of period 12 exactly, so it isolates $m(t)$ up to residual noise. For the boundary values and for the prediction we use the linear extrapolation $\hat{a} + \hat{b}\,t$ obtained by OLS on the centred moving average; we find

$$\hat{a} = 0.813, \qquad \hat{b} = 0.01027.$$

Figure 4 superposes X_t and $\hat{m}(t)$.

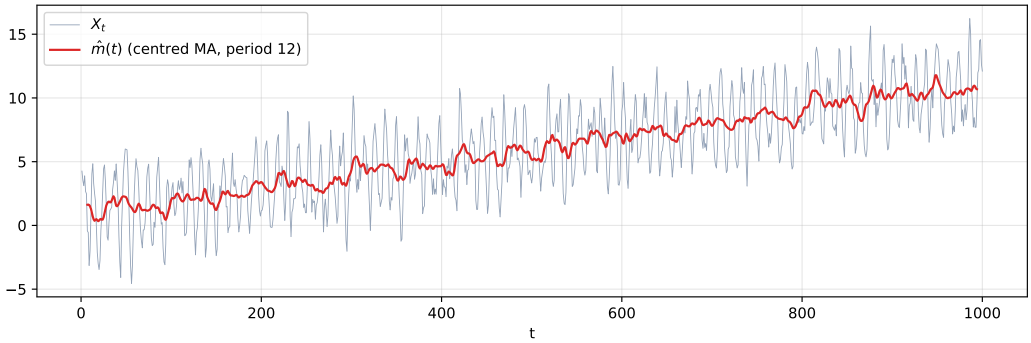


Figure 4: Raw series (grey) and centred moving-average trend estimate $\hat{m}(t)$.

Seasonality. With the detrended residuals $u_t = X_t - \hat{m}(t)$, we average per season and recentre to enforce $\sum_k \hat{s}(k) = 0$:

$$w_k = \frac{1}{n_k} \sum_{t \equiv k\,[12]} u_t, \qquad \hat{s}(k) = w_k - \tfrac{1}{12} \sum_{j=1}^{12} w_j.$$

The twelve coefficients are reported in Table 1 and plotted in Figure 5. The shape is a smooth sinusoidal pattern with a maximum near $k = 2$ and a minimum near $k = 8$.

k	1	2	3	4	5	6	7	8	9	10	11	12
$\hat{s}(k)$	2.72	2.85	2.12	0.65	−0.81	−2.01	−2.53	−2.55	−1.95	−0.86	0.58	1.79

Table 1: Estimated seasonal coefficients $\hat{s}(k)$.

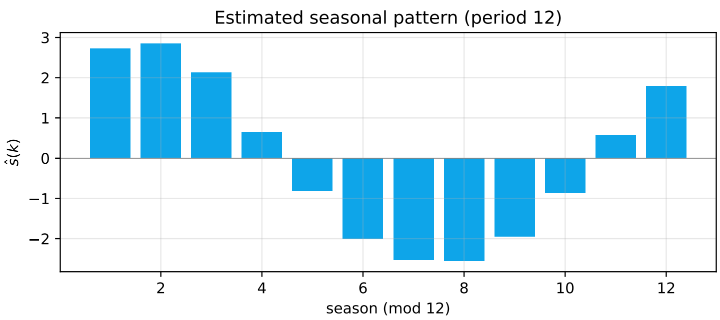
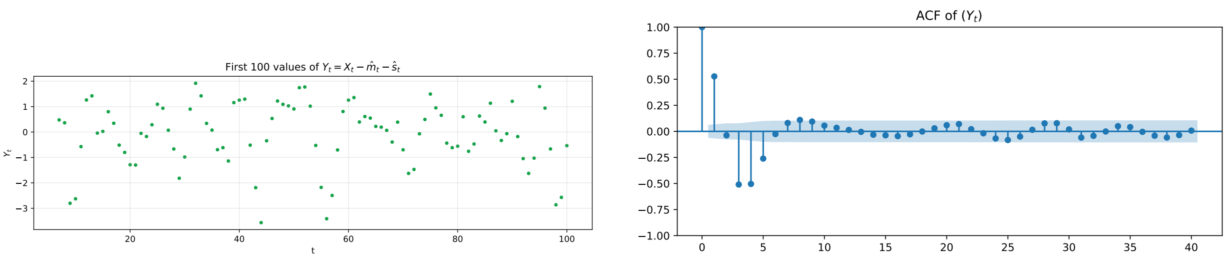


Figure 5: Estimated seasonal coefficients $\hat{s}(k)$, $k = 1, \ldots, 12$.

2.2 (a) Plot and (b) ACF of Y_t



(a) Y_t, $1 \le t \le 100$.

(b) ACF of Y_t.

Figure 6: Residual series $Y_t = X_t - \hat{m}_t - \hat{s}_t$.

The ACF drops immediately to near zero and stays in the confidence band, so we can assume that the series is stationary. We can check this with the Dickey–Fuller test:

- H_0: the series has a unit root (non-stationarity);
- H_1: the series is stationary.

We obtain ADF $= -12.61$, p-value $\approx 1.7 \cdot 10^{-23}$. The p-value is well below 0.05, so we reject H_0: Y_t is likely stationary, meaning that we can consider using an ARMA to estimate the series. This is consistent with what we observed from the ACF.

2.3 Choice of orders (p, q) for the ARMA

Criterion. We prefer the BIC over the AIC because the BIC is asymptotically consistent, whereas the AIC has a positive probability of selecting an overfitted model asymptotically.

Grid. We first searched on the small grid $p, q \in \{0, \dots, 5\}$, where the BIC minimum is ARMA$(2, 3)$ with BIC $= 2347.89$. To check that we are not missing a larger model we extended the grid to $p, q \in \{0, \dots, 15\}$; the minimum is then much lower and is attained at MA(10), with BIC $= 2297.98$ (Table 2). The BIC of MA(10) is clearly lower than that of ARMA$(2, 3)$, so we keep MA(10).

p	q	AIC	BIC
0	10	2239.23	**2297.98**
0	9	2244.44	2298.29
1	9	2244.07	2302.81
3	6	2249.99	2303.84
0	11	2250.76	2314.40
2	3	2313.62	2347.89
2	4	2311.18	2350.34
5	3	2302.02	2350.98

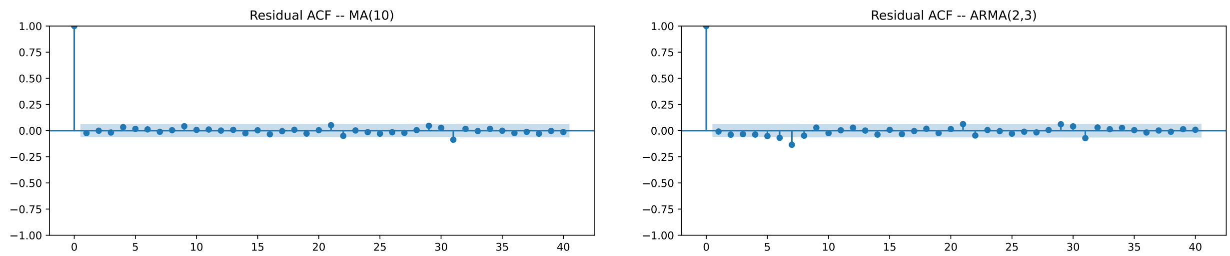
Table 2: Top of the BIC ranking for ARMA(p, q) models fitted to Y_t. The first five rows are taken from the full grid $\{0, \dots, 15\}^2$; the last three are the best of the small grid $\{0, \dots, 5\}^2$.

Fitted MA(10). Estimated model (all coefficients significant at the 5% level except $\hat{\theta}_{10}$, $p = 0.056$):

$$Y_t = \varepsilon_t + \sum_{k=1}^{10} \hat{\theta}_k \, \varepsilon_{t-k}, \qquad \hat{\sigma}^2 = 0.551,$$

with $\hat{\theta} = (0.32, -0.18, -0.90, -0.87, -0.36, 0.12, 0.31, 0.32, 0.20, 0.06)$.

Residual diagnostics. The residual ACF of MA(10) (Figure 7a) looks closer to white noise than that of ARMA$(2, 3)$ (Figure 7b). Almost all spikes of the MA(10) residuals are inside the confidence band, except at lags 21 and 31; for ARMA$(2, 3)$ many spikes lie outside the band. The MA(10) therefore captures the structure of Y_t better.



(a) MA(10).

(b) ARMA(2,3).

Figure 7: Residual ACFs.

We confirm this with a portmanteau test. The course covers the Box–Pierce statistic

$$Q_{\mathrm{BP}} = n \sum_{k=1}^{h} \hat{\rho}_k^2,$$

and we also report the Ljung–Box statistic, whose finite-sample distribution is closer to a χ_h^2 than is that of Box–Pierce.[1]

- H_0: $\rho_1 = \cdots = \rho_h = 0$ (residuals are white noise);
- H_1: there exists $j \in \{1, \ldots, h\}$ with $\rho_j \neq 0$.

	MA(10)		ARMA(2,3)	
lag h	Q_{LB}	p-value	Q_{LB}	p-value
1	0.54	0.46	0.07	0.79
4	1.94	0.75	4.07	0.40
6	2.36	0.88	11.37	0.08
7	2.48	0.93	29.49	**0.00012**
8	2.50	0.96	31.78	**0.00010**
12	4.45	0.97	33.97	**0.00068**
20	7.26	1.00	37.72	**0.00957**

Table 3: Ljung–Box statistic on the residuals. Bold entries are below 0.05 (reject white noise).

The table above shows the test statistics on the residuals of both models. For MA(10) all p-values are above 0.05; for ARMA$(2, 3)$ many p-values fall below 0.05 starting from lag 7.

Therefore we do *not* reject H_0 for MA(10): the residuals are white noise and the model is well specified. We do reject H_0 for ARMA$(2, 3)$: there is remaining autocorrelation, the model is misspecified. The residual ACF and the Ljung–Box / Box–Pierce tests both confirm that the MA(10) residuals are white noise.

2.4 Forecasts for $t = 1001, \ldots, 1100$

We extrapolate each component:

$$\widehat{X}_t = (\hat{a} + \hat{b}\, t) + \hat{s}\big((t-1) \bmod 12 + 1\big) + \widehat{Y}_t, \qquad t = 1001, \ldots, 1100,$$

where $\widehat{Y}_t$ is the conditional expectation from the MA(10). Note that while AR(p) models can propagate predictions arbitrarily far because the model depends on past observed values, an MA(q) has only q steps of memory: beyond the q-th step the conditional expectation collapses to the

[1] See https://en.wikipedia.org/wiki/Ljung-Box_test.

unconditional mean (here ≈ 0). Therefore $\widehat{Y}_t$ is informative only for the first few horizons; the bulk of the $\widehat{X}_t$ forecast is driven by the extrapolated trend $\hat{m}(t)$ and the seasonal component $\hat{s}(t)$.

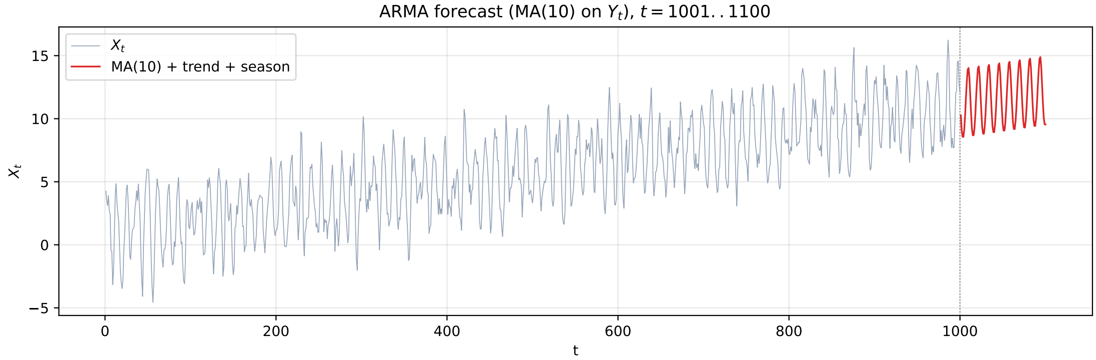


Figure 8: Forecast of X_t via the decomposition with MA(10) on Y_t.

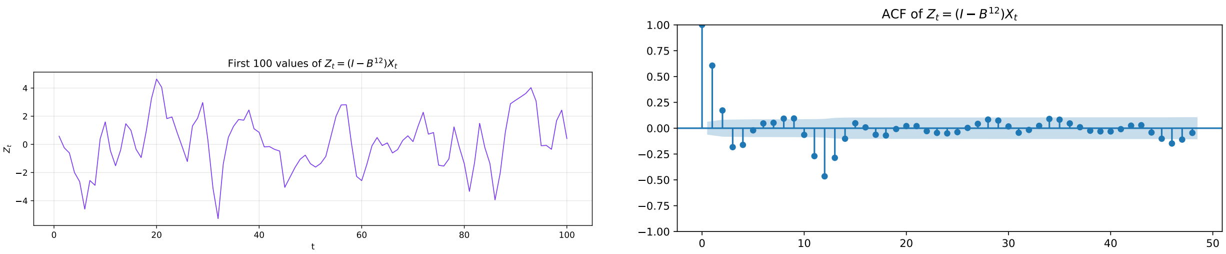
3 SARIMA modelling

3.1 Seasonal differentiation

We consider

$$Z_t = (I - B^{12})X_t = X_t - X_{t-12}, \qquad 13 \le t \le 1000,$$

of length $1000 - d = 988$. Figure 9a shows the first 100 values and Figure 9b the ACF.



(a) Z_t, first 100 values.

(b) ACF of Z_t.

Figure 9: Series after seasonal differentiation.

The ACF of Z_t decays quickly to zero and stays in the confidence band, with a single large negative spike at lag 12; the slow decay and periodic structure observed for X_t have disappeared. This single negative spike at the seasonal lag suggests an MA(1) seasonal component ($Q = 1$) in the SARIMA. The ADF test on Z_t gives ADF $= -9.94$, p-value $\approx 3 \cdot 10^{-17}$: we reject H_0 and conclude that Z_t is stationary. A single seasonal difference ($D = 1$) is therefore enough; no additional regular difference d is required.

3.2 Choice of the SARIMA orders

We search a SARIMA$(p, 0, q) \times (P, 1, Q)_{12}$ on the grid $(p, q) \in \{0, \dots, 3\}^2$ and $(P, Q) \in \{0, 1\}^2$ and minimise the BIC. The top of the ranking is in Table 4.

p	q	P	Q	AIC	BIC
3	3	0	1	2743.48	**2782.52**
3	3	1	1	2745.89	2789.81
2	2	0	1	2862.15	2891.43
2	3	0	1	2860.16	2894.32
3	2	0	1	2862.63	2896.79
0	3	0	1	2883.48	2907.88

Table 4: Top of the BIC ranking for the SARIMA grid.

The BIC is minimised by

$$\boxed{\mathrm{SARIMA}(3,0,3)\times(0,1,1)_{12}},$$

which is the model we retain. The decision is consistent with the shape of the seasonally-differenced ACF (single negative spike at lag 12 $\Rightarrow Q=1$) and with the observation that no additional non-seasonal difference is needed ($d=0$). The non-seasonal orders $p=q=3$ are picked up by the BIC; a model where the regular polynomials $\Phi(B)$ and $\Theta(B)$ are restricted to lower orders sacrifices a substantial amount of likelihood (+88 points of BIC for $p=q=2$).

Estimated model.

$$\Phi(B)\,(I-B^{12})X_t \;=\; \Theta(B)\,(1+\hat{\Theta}_1 B^{12})\,\varepsilon_t, \quad \hat{\sigma}^2 = 0.91,$$

with $\hat{\phi}=(1.477,\,-0.927,\,0.449)$, $\hat{\theta}=(-0.698,\,0.347,\,-0.649)$, $\hat{\Theta}_1=-1.00$. The autoregressive coefficients and the seasonal MA coefficient are highly significant ($|z|>10$, $|z|=2.5$ for the seasonal MA), confirming the relevance of those orders. The regular MA coefficients have large standard errors – their individual significance is poor – but the joint BIC clearly prefers the $(3,0,3)\times(0,1,1)_{12}$ specification over the $(3,0,0)$ or $(3,0,2)$ alternatives, which indicates that the three MA lags act jointly even if they are not individually well identified.

Residual diagnostics. The residual ACF (Figure 10) is essentially flat inside the confidence band. The Ljung–Box / Box–Pierce tests (Table 5) do not reject the white-noise hypothesis at any lag up to 20 (all p-values >0.9). The model is therefore well specified.

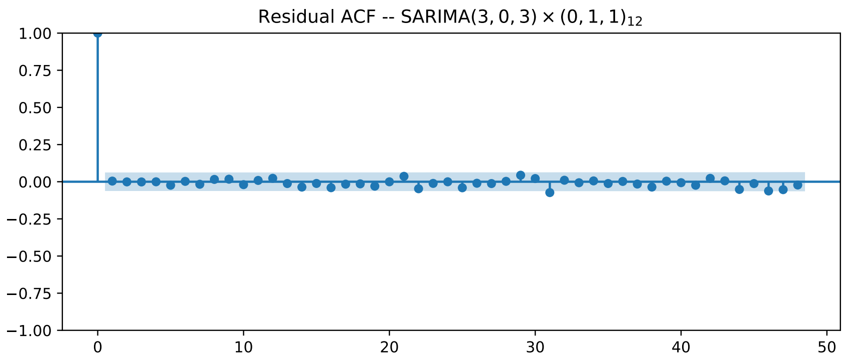


Figure 10: Residual ACF of $\mathrm{SARIMA}(3,0,3)\times(0,1,1)_{12}$.

lag h	Q_{LB}	p-value (LB)	p-value (BP)
1	0.02	0.89	0.89
4	0.02	1.00	1.00
8	1.12	1.00	1.00
12	2.40	1.00	1.00
20	–	1.00	1.00

Table 5: Portmanteau tests on the SARIMA residuals.

3.3 Forecasts for $t = 1001, \ldots, 1100$

The SARIMA forecast is the conditional expectation in the state-space representation; an associated 95 % predictive interval is available. The result is shown in Figure 11 and the 100 values are written in `predictions_sarima.txt`.

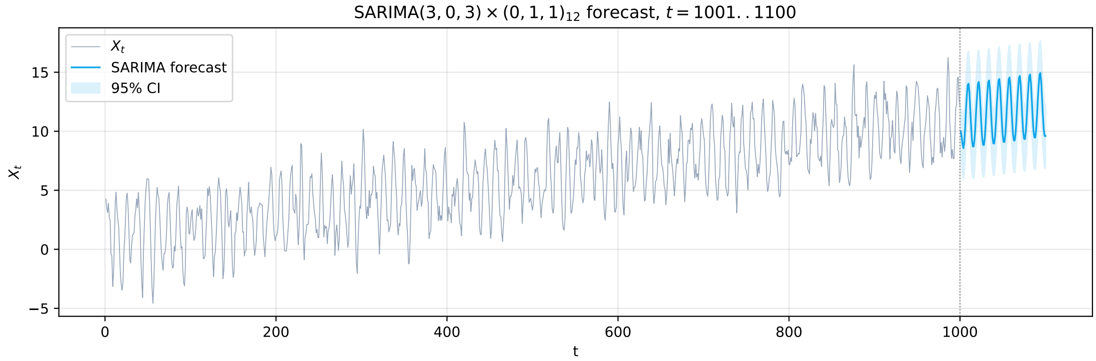


Figure 11: Forecast and 95 % interval from SARIMA$(3,0,3) \times (0,1,1)_{12}$, $t = 1001, \ldots, 1100$.

4 Comparison and conclusion

Figure 12 overlays the two predicted trajectories.

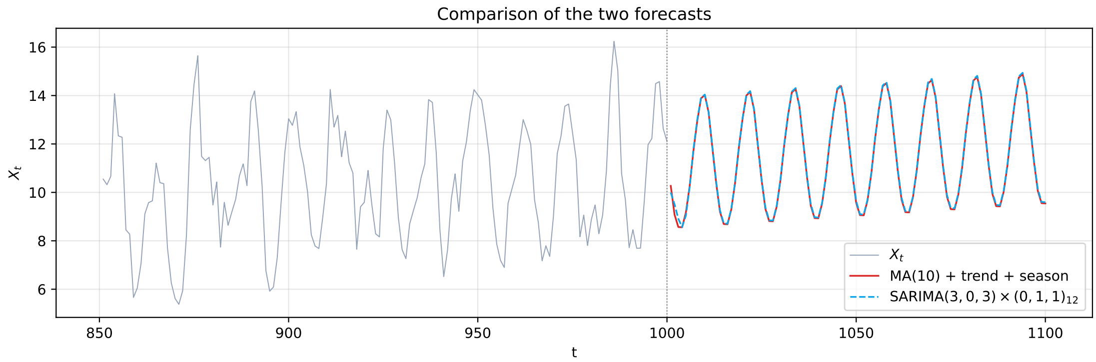


Figure 12: Forecasts from the MA(10) decomposition and from the SARIMA model.

Model	log-likelihood	AIC	BIC
MA(10) on Y_t	-1107.61	2239.23	2297.98
SARIMA$(3,0,3) \times (0,1,1)_{12}$ on X_t	-1363.74	2743.48	2782.52

Remark: *The two BIC values cannot be compared directly*: the MA(10) is fitted to the residual series Y_t, whereas the SARIMA is fitted to the full X_t. The MA(10) only needs to explain the leftover noise after the trend and the seasonality have been removed by deterministic estimators, so it carries a much lower likelihood scale by construction.

Qualitative comparison. Both models pass every diagnostic:

- residual ACFs lie within the confidence band;
- Ljung–Box / Box–Pierce do not reject the white-noise hypothesis at any lag up to 20.

The two forecasts have a very similar shape over the next 100 steps (Figure 12): both reproduce the seasonal pattern, and both follow an almost-linear upward drift.

Which model is preferable? We retain the **SARIMA**$(3, 0, 3) \times (0, 1, 1)_{12}$ as our reference model, for several reasons:

1. **The SARIMA does not propagate the estimation error of $\hat{m}$ and $\hat{s}$.** Our ARMA-based approach is really a two-stage procedure: we first estimate the trend and the seasonality, and only *then* do we fit an ARMA on what is left. Any error we made on $\hat{m}$ or $\hat{s}$ (and there are several potential sources – boundary effects of the centred moving average, the fact that we forced the trend to be linear, etc.) ends up baked into Y_t and is carried over to the forecast as-is. The SARIMA avoids this entirely: trend and seasonality are handled inside the model itself, via the differencing operator $(I - B^{12})$.

2. **The SARIMA keeps producing meaningful uncertainty at long horizons.** An MA(q) only remembers q steps into the past, so once we go past the q-th step ahead the conditional expectation of Y_t collapses to its (zero) mean. Concretely, after lag 10 our $\widehat{X}_t$ forecast is just $\hat{a} + \hat{b}\,t + \hat{s}(t)$ – a perfectly deterministic line plus a periodic pattern, with no more residual variability. The SARIMA, on the contrary, keeps a genuine predictive distribution at every horizon, which is much closer to what we would actually expect from the data.

3. **The SARIMA is more forgiving if our assumptions about m and s are slightly off.** The brief assumes that the trend is exactly linear and the seasonality is exactly deterministic and of period 12. In practice neither of these is ever perfectly true; the amplitude of the seasonal pattern might drift slowly, or the trend might curve a little. The decomposition model would carry such a misspecification as a permanent bias, whereas the seasonal differencing of the SARIMA absorbs slow variations naturally.

4. **The BIC backs up the choice.** Among all the SARIMA candidates we tried, $(3, 0, 3) \times (0, 1, 1)_{12}$ is the one with the lowest BIC, and the margin is not tight – the next best model is 7 BIC points behind.

That said, the MA(10) decomposition has interpretive value: it makes the trend and seasonal components explicit, and its short-term forecast is very close to that of the SARIMA